A4.9 Usage examples

This section gives examples of memory type usage.

A4.9.1 Use of Device memory types

The specification supports the combined use of Device Non-buffered and Device Buffered memory types to force write transactions to reach their final destination and ensure that the issuing master knows when the transaction is visible to all other masters.

A write transaction that is marked as Device Buffered is required to reach its final destination in a timely manner. However, the write response for the transaction can be signaled by an intermediate buffer. Therefore, the issuing master cannot know when the write is visible to all other masters.

If a master issues a Device Buffered write transaction, or stream of write transactions, followed by a Device Non-buffered write transaction, and all transactions use the same AXI ID, the AXI ordering requirements force all of the Device Buffered write transactions to reach the final destination before a response is given to the Device Non-buffered transaction. Therefore, the response to the Device Non-buffered transaction indicates that all the transactions are visible to all masters.

Note

A Device Non-buffered transaction can only guarantee the completion of Device Buffered transactions that are issued with the same ID, and are to the same slave device.

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Chapter A5
Multiple Transactions

This chapter describes the mechanism that enables out-of-order transaction completion and the issuing of multiple outstanding addresses. It contains the following sections:

- *AXI transaction identifiers* on page A5-76
- *Transaction ID* on page A5-77
- *Transaction ordering* on page A5-78
- *Removal of write interleaving support* on page A5-81.

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